MICHAELMAS 2026

AI for Financial Practitioners

Three lectures. Twelve experiments.

A progressive journey from knowledge and reasoning to autonomous systems, governed agency, swarms, and adaptive intelligence.

ALEJANDRO REYNOSO

Class of 2027

Course at a Glance



A Progressive Journey in Capability and Context

PRACTICE GIVES PURPOSE. SCIENCE PROVIDES CAPABILITY.

LECTURE 1

Institutional Perspective

From knowledge to action

- Make knowledge explicit, grounded, and inspectable.
- Turn evidence into structured professional judgment.

LECTURE 2

Technological Perspective

From action to governed agency

- Observe, decide, act, and learn through feedback.
- Separate system capability from delegated authority.

LECTURE 3

Adaptive Perspective

From governed agency to adaptive intelligence

- Coordinate specialized agents around complex tasks.
- Let organizations adapt under explicit governance.

How Each Three-Hour Lecture Works

Opening reading and reflection	20 min
Conceptual bridge	15 min
Application block A: two examples	60 min
Break	10 min
Application block B: two examples	60 min
Synthesis and discussion	15 min
Total	**180 min**

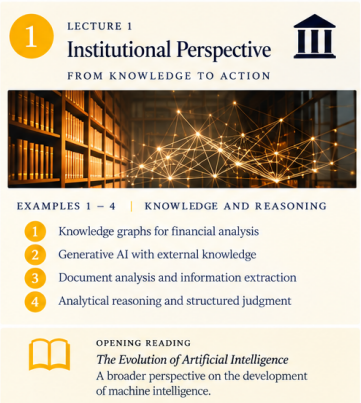
PEDAGOGICAL LOGIC

- A reflective reading opens the intellectual question.
- Four experiments make the architecture tangible.
- A final synthesis reconnects technology to finance.

Across the course

6 application blocks **12** experiments

Lecture 1 — Institutional Perspective



FROM KNOWLEDGE TO ACTION

- **Structure knowledge:** relationships among facts, documents, concepts, institutions, and evidence.
- **Ground answers:** connect generative AI to external sources and inspect provenance.
- **Make judgment reviewable:** move from fluent answers to a defensible path from evidence to conclusion.

Lecture 1 — Experiments 1 and 2

KNOWLEDGE AND REASONING

1

Knowledge graphs for financial analysis

Represent entities, documents, events, and concepts as a navigable structure. The central question becomes: what path of evidence supports the conclusion?

Professional lens: regulatory research, institutional memory, due diligence and compliance.

2

Generative AI with external knowledge

Connect generative models to explicit external sources so reasoning can be grounded, refreshed, and checked rather than confined to model memory.

Professional lens: research workflows, policy analysis and investment intelligence.

Lecture 1 — Experiments 3 and 4

FROM DOCUMENTS TO STRUCTURED JUDGMENT

3

Document analysis and information extraction

Turn unstructured financial and legal documents into organized evidence: entities, obligations, events, tables, and relationships that can be reviewed systematically.

Professional lens: credit files, legal documents and disclosure analysis.

4

Analytical reasoning and structured judgment

Move beyond extraction to explicit comparison, evaluation, scoring, and decision logic. Assumptions and reasoning steps should be visible enough to challenge.

Professional lens: investment committees, risk assessment and governance.

Lecture 1 — What Changes for the Practitioner?

FROM CASUAL USE

“Give me an answer.”

- Prompt a model.
- Read a plausible response.
- Rely heavily on conversational fluency.

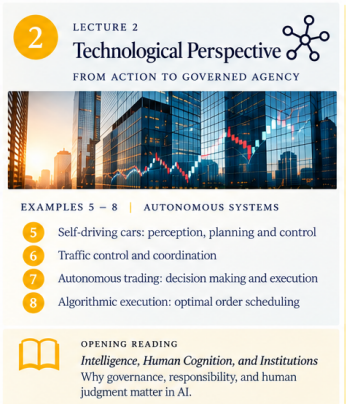
TOWARD PROFESSIONAL USE

“Show me the evidence and the reasoning path.”

- Structure knowledge and evidence.
- Inspect provenance and limitations.
- Make judgment explicit and reviewable.

Knowledge becomes useful when it can support accountable action.

Lecture 2 — Technological Perspective



FROM ACTION TO GOVERNED AGENCY

- **Autonomy is a loop:** observe, interpret, decide, act, and learn from feedback.
- **Control is part of intelligence:** planning, execution, state estimation, and intervention matter as much as reasoning.
- **Finance raises the stakes:** permissions, risk limits, validation, escalation, and human decision rights become architectural requirements.

Lecture 2 — Experiments 5 and 6

AUTONOMOUS SYSTEMS: PERCEIVE, PLAN, COORDINATE

5

Self-driving cars: perception, planning and control

Use the self-driving system as a transparent model of autonomy: sense, estimate state, plan, act, and continuously correct under safety constraints.

Professional lens: a template for financial systems that act under uncertainty.

6

Traffic control and coordination

Shift from optimizing one actor to coordinating many. Feedback, congestion, local objectives, and system-wide performance turn autonomy into a coordination problem.

Professional lens: market infrastructure, liquidity and operations control.

Lecture 2 — Experiments 7 and 8

AUTONOMY IN FINANCIAL EXECUTION

7

Autonomous trading: decision making and execution

Translate the observe–decide–act–learn loop into markets. Strategy, position sizing, risk limits, execution permissions, feedback, and stop conditions form one architecture.

Professional lens: algorithmic execution, market monitoring and supervised autonomy.

8

Algorithmic execution: optimal order scheduling

Focus autonomy on order scheduling: balance market impact, timing risk, liquidity, and explicit execution constraints.

Professional lens: execution quality, transaction costs and auditable automation.

Lecture 2 — Capability Is Not Authority

THE AUTONOMOUS LOOP

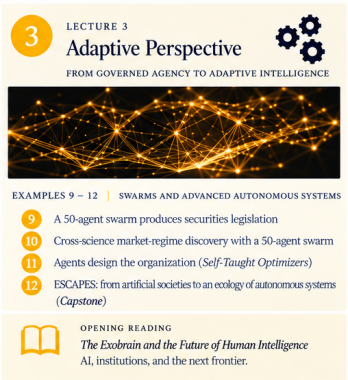
1. **Perceive** the relevant state.
2. **Judge** what action is warranted.
3. **Act** within permissions and constraints.
4. **Observe feedback** and update the decision.

THE INSTITUTIONAL HARNESS

- Controlled data and tools.
- Deterministic checks and validation.
- Permissions, limits, escalation and stop conditions.
- Provenance, monitoring and human decision rights.

Stronger intelligence makes the design of governance more consequential—not less.

Lecture 3 — Adaptive Perspective



FROM GOVERNED AGENCY TO ADAPTIVE INTELLIGENCE

- **Swarms differentiate expertise:** roles, common evidence, proposals, critique, synthesis, and validation.
- **Architecture can adapt:** agents help determine which specialists are needed and how teams coordinate.
- **The frontier becomes organizational:** intelligence may reside in governed ecologies of autonomous systems.

Lecture 3 — Experiments 9 and 10

GOVERNED SWARMS FOR PRODUCTION AND DISCOVERY

9

A 50-agent swarm produces securities legislation

Organize an auditable workflow from diagnosis and specialized proposals through adversarial review, synthesis, deterministic checks, and a reform package for human legal review.

Professional lens: capital-markets policy, regulation and high-stakes multidisciplinary work.

10

Cross-science market-regime discovery with a 50-agent swarm

Let specialists from multiple sciences propose competing hypotheses about latent market regimes, reconcile evidence, and test executable models for investment research.

Professional lens: regime detection, portfolio risk and research under uncertainty.

Lecture 3 — Experiments 11 and 12

ARCHITECTURES THAT ADAPT

11

Agents design the organization — Self-Taught Optimizers

Explore systems in which agents help determine their own working architecture: specialties, constellations, responsibilities, and procedures can themselves be evaluated and improved.

Professional lens: adaptive research, dynamic teams and governed self-improvement.

12

ESCAPES — from artificial societies to an ecology of autonomous systems

The capstone links separate autonomous societies through bounded informational and relational "escapes," negotiated compacts, interoperability, and institutional learning.

Professional lens: knowledge transfer, organizational learning and auditable autonomy.

Lecture 3 — The Frontier Moves from Models to Institutions

INDIVIDUAL INTELLIGENCE

One capable model

Useful, but limited by one context, one scaffold, and one working perspective.

COLLECTIVE INTELLIGENCE

Many specialized agents

Roles, common evidence, debate, synthesis, validation, and governance.

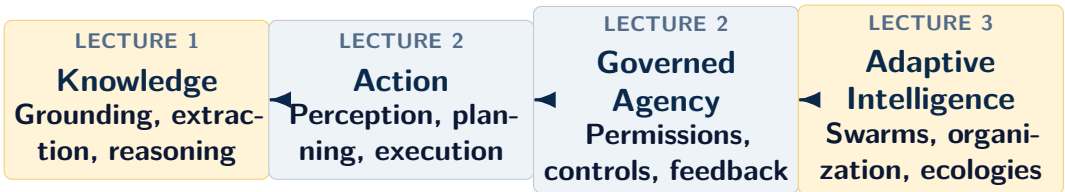
ADAPTIVE INTELLIGENCE

Organizations that change

Teams, procedures, boundaries, and institutional relationships adapt under explicit rules.

The design problem becomes: how should intelligence be organized?

The Ladder You Will Have Climbed



The purpose is not to memorize a catalogue of technologies. The practical outcome is an **architectural repertoire**: the ability to recognize what form of intelligence a financial problem requires, choose methods appropriate to its complexity, and resist using a more elaborate system when a simpler one is sufficient.

Learning Without Technical Prerequisites

NO CODING SKILLS OR ADVANCED MATHEMATICS ARE REQUIRED TO FOLLOW THE COURSE.

01 Monographs

Explain the problem, conceptual architecture, experiment, and lessons in accessible written form.

02 Lecture slides

Organize the classroom narrative with diagrams, visual interpretations, and conceptual bridges.

03 Colab notebooks

Preserve the underlying computational experiments for students who wish to investigate further.

Classroom method. We use diagrams, selected notebook cells, simulated outputs, and guided visualizations. The recurring questions are: *What does the system know? How does it decide? What is it allowed to do? How can it fail? How should its outputs be evaluated?*

What You Will Take Away

THREE KINDS OF SITUATIONAL AWARENESS

- **Scientific:** distinguish established capability from exploratory research.
- **Institutional:** recognize the gap between a convincing demonstration and a deployable financial system.
- **Professional:** ask the right questions about architecture, validation, autonomy, risk, and human judgment.

FROM USER TO INSTITUTIONAL PRACTITIONER

- Define the problem before selecting the tool or architecture.
- Structure reliable workflows around evidence, controls, and evaluation.
- Understand permissions, oversight, auditability, resilience, cybersecurity, and accountability.

The goal is fluency in the architectures, limits, and responsibilities of AI in finance.

THE LASTING QUESTION

What form of intelligence does this problem require, how should we build and govern it, and what will its use mean for the institutions and people who depend upon it?